Feasibility of Sparse Autoencoders to Identify Interpretable Features from Protein Language Models' Latent Representation

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Generative protein-design models produce candidate binders without revealing what internally distinguishes a strong binder from a weak one, limiting rational improvement and failure diagnosis. We apply sparse autoencoders (SAEs), a mechanistic-interpretability technique, to residue-level ESM-C activations of previously Boltz-designed binder candidates against Vilip-1, a blood biomarker of acute neurological injury, to identify human-interpretable features captured in these underlying latent representations without generating or modifying any binder sequences ourselves. Compared to leveraging solely synthetic designs, mixing real, evolutionarily distinct natural-protein sequences into SAE training was the largest lever for generalization found, raising held-out natural-protein reconstruction fidelity from 0.39 to 0.60 fraction of variance explained. To test whether learned features reflect genuine protein structure rather than training artifacts, we compared two independently trained SAE dictionaries: encoding binders alone and encoding binders with full cross-attention to their target before target positions were discarded. Despite using different encoding, each dictionary's most generic features converge on the same real-protein residues at a rate statistically inconsistent with chance (hypergeometric test against the $\sim$7M-residue candidate space, $p \approx 7 \times 10^{-28}$), indicating that cross-attention preserves the learned signal. This convergence survives a shared-seed confound check (5 of 6 agreeing cases use different learned feature indices per dictionary) and is independently corroborated by InterPro domain annotations and calibrated LLM-assisted labeling that withholds a description when evidence is scattered. Together, these results establish that SAE features recovered from protein language model activations carry real, reproducible structural signal, a necessary validation step before any attempt to use such features to inform or steer binder design.